SPECIFICATION TITLE	SPEC NUMBER	REVISION	EFFECTIVE DATE
MATERIAL REQUIREMENTS FOR FORGINGS, BARS AND MILL SHAPES	**CMS-002**	**L**	**10/18/2024**

Scope	To establish the Cactus Wellhead material requirements for forgings and wrought products such as tubulars, bars and mill shapes. Suppliers shall use this specification in conjunction with the specific Cactus Wellhead Material Specification (CMS) or an approved supplier material specification. The Cactus Wellhead Material Specification shall take precedence over any conflict between this document and the CMS. For API Spec 6A PSL-3, 75K yield strength bodies, weighing over 1,000 lbs. (454 kg) during heat treating refer to CMS-002-1.
References	• API Specification 6A, Specification for Wellhead and Tree Equipment • ASTM A370, Standard Test Methods and Definitions for Mechanical Testing of Steel Products • ASTM A751, Standard Test Method, Practices & Terminology for Chemical Analysis of Steel Raw Materials • ASTM E10, Standard Test Method for Brinell Hardness of Metallic Materials • ASTM E1086, Standard Test Method for Analysis of Austenitic Stainless Steel by Spark Atomic Emission Spectrometry • ASTM E110, Standard Test Method for Indentation Hardness of Metallic Materials by Portable Hardness Testers • ASTM E112, Standard Test Methods for Determining Average Grain Size • ASTM E18, Standard Test Methods for Rockwell Hardness of Metallic Materials • ASTM E415, Standard Test Method for Analysis of Carbon and Low-Alloy Steel by Spark Atomic Emission Spectrometry • ASTM E92, Standard Test Method for Vickers Hardness of Metallic Materials • Form CMS-002, Cactus Wellhead Material Certification Report – Forgings, Bars, and Mill Shapes • ISO 14284, Steel and iron. Sampling and preparation of samples for the determination of chemical composition • SAE AMS2750F, Pyrometry • SAE AMSH6875, Process for Heat Treatment of Steel • ASTM E562, Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count **Note:** For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies, except that API reference standards may be used upon issue and shall become mandatory upon the effective date specified by the publisher or 12 months from the date of the revision (where no effective date is specified).

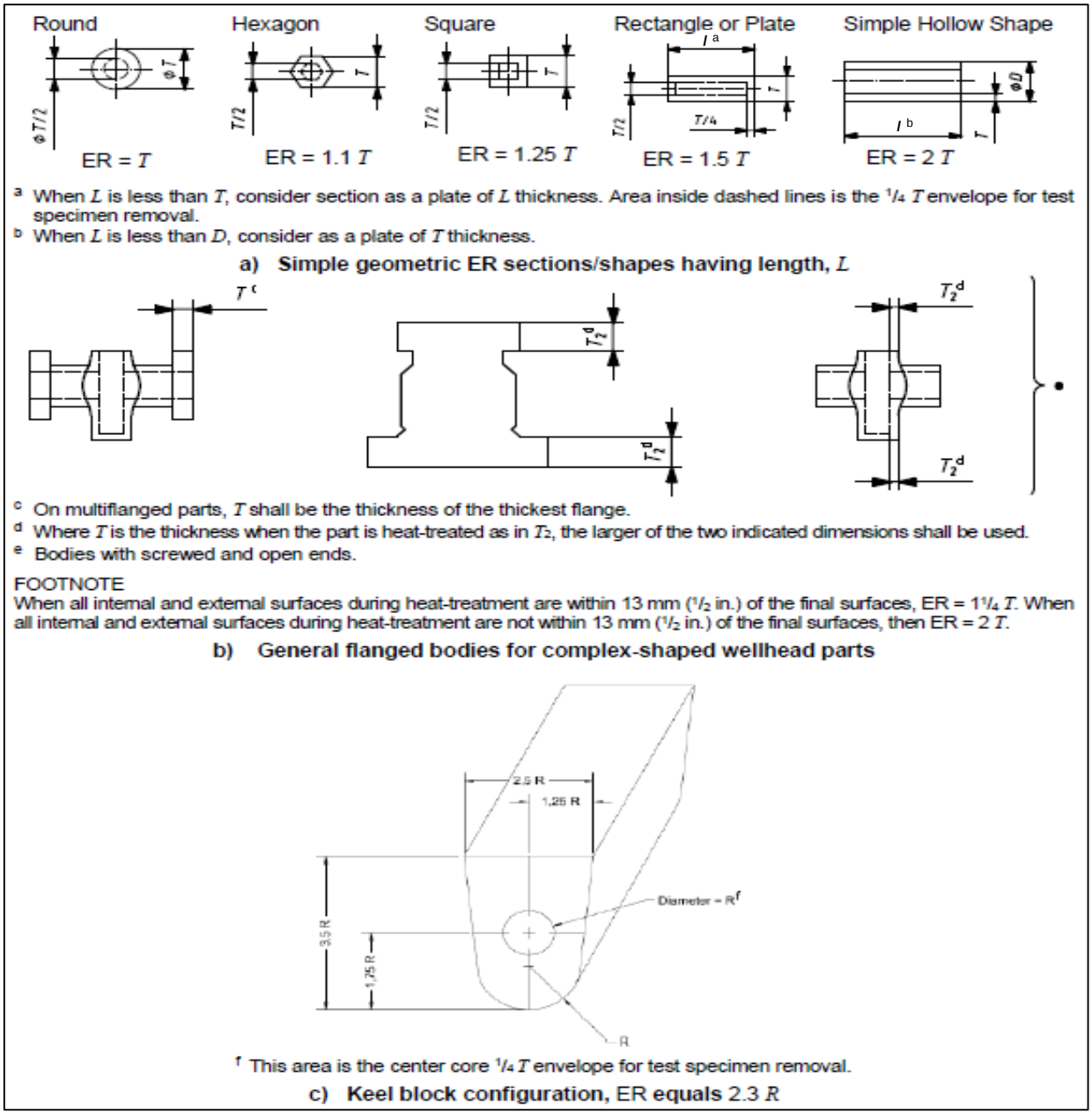
<table>
<tr><td>Approved Testing Methods</td><td>Suppliers shall determine chemistry, mechanicals, and impact test results in accordance with the latest edition of the following nationally or internationally recognized standards listed below.</td></tr>
<tr><td>Approved Testing Methods</td><td>Suppliers shall certify to the testing method standards in affect at the time of actual material testing.
<table>
<tr><th>Test Method</th><th>Approved Testing Standards</th></tr>
<tr><td>Chemical Composition Analysis Specified and approved by Cactus Wellhead</td><td>• ASTM A751 - Standard Test Methods, Practices, and Terminology for Chemical Analysis of Steel Products. All test methods listed in ASTM A751 Section 2, are acceptable for chemical analysis of steels, such as
• ASTM E415 - Standard Test Method for Analysis of Carbon and Low-Alloy Steel by Spark Atomic Emission Spectrometry
• ASTM E1086 - Standard Test Method for Analysis of Austenitic Stainless Steel by Spark Atomic Emission Spectrometry
• ISO 14284 - Steel and iron. Sampling and preparation of samples for the determination of chemical composition</td></tr>
<tr><td>Mechanical Properties</td><td>ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products</td></tr>
<tr><td>Impact Testing</td><td>ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products</td></tr>
<tr><td>Hardness Testing</td><td>• ASTM E10 - Standard Test Method for Brinell Hardness of Metallic Materials
• ASTM E18 - Standard Test Method for Rockwell Hardness of Metallic Materials
• ASTM E92 – Standard Test Method for Vickers Hardness of Metallic Materials
• ASTM E110, Standard Test Method for Indentation Hardness of Metallic Materials by Portable Hardness Testers</td></tr>
<tr><td>Grain Testing</td><td>ASTM E112 – Standard Test Methods for Determining Average Grain Size
ASTM E562 - Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count</td></tr>
</table></td></tr>
<tr><td>Material Processing</td><td>Hot Working Practices – Wrought materials shall be formed using a documented hot-working practice that produces a wrought structure throughout with a minimum 3:1 reduction ratio.
CHINA CLOSED/OPEN DIE FORGINGS SHALL HAVE A MINIMUM 5:1 REDUCTION RATIO.
Steel Mill Practice - Material shall be pressure vessel quality and free from defects such as laps, slag, porosity, scale, cracks, and fissures.</td></tr>
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	Melting Practice Carbon and Low Alloy Steel - Electric furnace (EF) or basic oxygen furnace shall be acceptable steel melting practices. Any additional processing such as argon oxygen decarburization (AOD), electroslag re-melting (ESR), ladle refining, vacuum degassing, or vacuum arc re-melting (VAR) maybe be performed when required. **Melting Practice Austenitic Stainless** Steel – Two-step process consisting of electric furnace (EF) melting or vacuum induction melting followed by argon oxygen decarburization (AOD), vacuum degassing, vacuum arc re-melting (VAT) or electroslag re-melting (ESR). **Grain size for Low Alloy Steels -** Grain size of 5 or finer (i.e. 5 or higher number) per ASTM E112 is required for low alloy steels. A statement from the mill of 'fine grain practice" may be used in lieu of the actual number. **Radioactivity Free Material -** Raw material melt sources shall have procedures in place to assure that the raw material used for bars and forgings is free from radioactivity.
Heat Treating	**Equipment Qualification:** Heat treatment operations shall be performed with heat-treating equipment that satisfies one of the following; • Calibrated in accordance with API 6A section 6.5.2 and 6.5.3 and surveyed in accordance with API 6A Annex M and section 6.5.4.1 for batch type furnaces. Or; • Equipment has been calibrated and surveyed in accordance with SAE AMS 2750 Class 1, 2 or 3 for tempering and Class 1, 2, 3, 4 or 5 for normalizing and austenitizing. Furnaces used for continuous heat treatment shall be operated, maintained, modified and repaired in accordance with SAE AMS2750 or SAE AMSH6875 and calibrated/surveyed in accordance with API Specification 6A Annex M. **Method for Batch Heat Treatment:** The QTC shall experience the same heat treatment processing as the material/part(s) it qualifies. Time at temperature and heat treatment cycles shall comply with the specific CMS. If the QTC is not heat treated as part of the same heat treatment lot as the material/part(s) it qualifies, the austenitizing, solution treating, or age hardening (as applicable) temperatures for the QTC shall be within 25°F (14°C) of those for the material/part(s). The tempering temperature of the material/part(s) shall not be lower than 25°F (14°C) below that of the QTC. The upper limit shall be no higher than that permitted by the specific CMS. The cycle time of the QTC at each temperature shall not exceed that for the material/part(s). Oven shall be loaded with appropriate spacing between parts to ensure proper heat treatment. In no case shall parts be stacked or touching during heat treatment. **CHINA CLOSED/OPEN DIE FORGINGS** MUST BE LOADED EVENLY WITH MINIMUM OF 4" SPACING BETWEEN ALL PARTS. A THERMOCOUPLE SHALL BE PLACED ON THE SURFACE OF THE CENTER PRODUCTION

	PART AND CHARTED WITH THE OVEN THERMOCOUPLES TO VALIDATE PRODUCTION PARTS REACHED SOAK TEMP AND TIMES. **Quenching (for Quench and Tempered Materials):** For materials that are quenched and tempered, the temperature of the water or other quench media (such as oil or polymer) used to approximate the cooling rate of water at the start and completion of the quench, shall comply with the specific CMS. **Method for continuous furnace:** The QTC shall be from the same heat and heat treat lot as the material/part(s) it qualifies.
Chemical Composition	An analysis of each heat of steel shall be made by the supplier, preferably from a ladle sample taken at or near the time of pouring. The required chemical elements shall be reported in weight percent. While reporting of residual elements is not required, unless called out in the specific CMS, the supplier must ensure the total residuals do not exceed 1%. Chemical composition including composition tolerances are stated in the specific CMS.
Qualification Test Coupon Qualification	**Continuous furnace:** For material heat treated in a continuous furnace, the QTC shall consist of a sacrificial production part or a prolongation removed from the production part, the sacrificial production part or prolongation QTC shall qualify only production part(s) having an identical size and shape. The QTC shall qualify only material/part(s) produced from the same heat and heat treat lot. **Batch-type furnace:** For batch heat treatment only, if the QTC is a trepanned core or prolongation removed from a production part, the QTC may qualify only production part(s) having the same or smaller ER. The QTC shall qualify only material/part(s) produced from the same heat. **Production Parts:** • A sacrificial production part may be usable depending on the size and location of the test specimen from which it is selected. • For a production part with a complex shape or varying cross-section, it is not necessary that the prolongation be an extension of the largest cross-section of the part, but the extension shall meet or exceed the minimum ER required for a separate QTC. **Prolongations:** • When a prolongation is used, it shall remain integrally attached during all heat-treatment operations, except post-weld heat-treatment, stress relief, and any re-tempering or re-aging that can be required. • For tubular parts, bar stock, mill shapes, and other raw material with a uniform cross-section, the prolongation shall have the same cross-section as that of the raw material In no case shall the QTC be processed using a melting practice cleaner than that of the material it qualifies.

	QTC hot work ratios must be equal to or less than those used in processing the production material/part(s) it qualifies. The total hot work ratio for the QTC shall not exceed the total hot work ratio of the material/part(s) it qualifies.
China Closed/Open Die Forging Coupon Requirements	**CHINA CLOSED/OPEN DIE FORGINGS** MUST HAVE 3 INDIVIDUAL TEST COUPONS/PROLONGATIONS PER HEAT LOT. THESE SHALL BE EQUALLY SPACED IN THE OVEN WITH ONE BEING IN THE CENTER. ONE ON THE END WILL BE FOR FORGING SUPPLIER TO TEST, THE CENTER WILL BE CACTUS SUZHOU TO TEST AND THE THIRD WILL BE SENT TO CACTUS WELLHEAD IN BOSSIER CITY. FORGING SHAPES SHALL NOT HAVE ANY SHARP CORNERS PRIOR TO HEAT TREATING. ANY ROUGH MACHINING PRIOR TO HEAT TREATING MUST HAVE 3/8 – ½ IN. (9.5 – 13MM) RADIUS AT ALL CORNERS, TRANSITIONS AND INTERSECTIONS.
Qualification Test Coupon Equivalent Round	When mechanical testing is required by the CMS or the Engineering Product Specification (EPS), the test specimens shall be removed from the QTC after the QTC final heat treatment cycle. The size of a QTC for a part shall be determined using the Equivalent Round (ER) methods specified in API 6A below. Basic models for determining the ER of simple solid and hollow parts and more complicated parts shall apply and illustrated in Figure 1 below taken from API 6A. The ER of a studded type part shall be determined by using a thickness, *T*, equal to that of the thickest flange of that part. An ER determination for these parts shall be in accordance with the methods for complex-shaped parts shown in Figure 1. The ER of the QTC shall be equal to or greater than the dimensions of the material/part(s) it qualifies. The QTC shall be a minimum of 7" long for a standard specimen. The maximum ER required shall be 5" (4 x 4 QTC) unless otherwise specified in the specific CMS.

Figure 1 – Equivalent Round Models from API Spec 6A for reference



[a] When L is less than T, consider section as a plate of L thickness. Area inside dashed lines is the 1/4 T envelope for test specimen removal.
[b] When L is less than D, consider as a plate of T thickness.

a) Simple geometric ER sections/shapes having length, L

[c] On multiflanged parts, T shall be the thickness of the thickest flange.
[d] Where T is the thickness when the part is heat-treated as in T_2, the larger of the two indicated dimensions shall be used.
[e] Bodies with screwed and open ends.

FOOTNOTE
When all internal and external surfaces during heat-treatment are within 13 mm (1/2 in.) of the final surfaces, ER = 1 1/4 T. When all internal and external surfaces during heat-treatment are not within 13 mm (1/2 in.) of the final surfaces, then ER = 2 T.

b) General flanged bodies for complex-shaped wellhead parts

[f] This area is the center core 1/4 T envelope for test specimen removal.

c) Keel block configuration, ER equals 2.3 R

Tensile and Impact Test Specimens	When mechanical testing is required by the CMS or the Engineering Product Specification (EPS), the test specimens shall be removed from a QTC after the final QTC heat treatment cycle. **Tensile Test Specimens** shall be removed from the QTC such that the tensile specimen gauge length is at least 1/4 T from the ends of the QTC. Tensile test specimens shall be removed from the QTC such that their longitudinal centerline axis is wholly within the center core 1/4 T envelope for a solid QTC or within 1/8 inch (3mm) of the mid-thickness of the thickest section of a hollow QTC. If a

sacrificial production part(s) is used as a QTC, the test specimens shall be removed from the 1/4 T envelope location of the thickest section.

Impact Test Specimens shall be removed from the QTC such that the Charpy V-notch root is at least 1/4 T from the ends of the QTC. The impact specimens shall be removed such that the notch is within the 1/4 T envelope. If a sacrificial production part(s) is used as a QTC, the test specimens shall be removed from the 1/4 T envelope location of the thickest section.

CHINA CLOSED/OPEN DIE FORGINGS SHALL HAVE THREE SETS OF IMPACT TEST SPECIMENS REMOVED FROM EVERY QTC/PROLONGATION.

Tensile tests shall be conducted per ASTM A370. Standard size 0.500" (12.5 mm) diameter tensile specimens shall be used unless the physical configuration of the QTC prevents their use. In this case, the standard sub-size no smaller than 0.250" (6.25 mm) specimens referenced in ASTM A370 may be used.

- Tensile tests shall be performed at a temperature between 40° F and 120° F (4° C and 50° C) in accordance with the procedures specified in ASTM A370.
- Perform a minimum of one tensile test
- Results of tests shall satisfy the specific CMS
- If the results of the tensile test are not satisfactory, two additional tests on two additional test specimens (removed from the same QTC with no additional heat treatment) may be performed to qualify the material. The results of each of these tests shall satisfy the applicable requirements.

Impact tests shall be conducted per ASTM A370 using the Charpy V notch technique. Standard-sized impact specimens, 10 mm x 10 mm cross-section, shall be used, except where there is insufficient material, in which case the next smaller standard-size specimen obtainable shall be used. When sub size specimens are used an adjustment factor shall be applicable per the Sub Size Specimens section below.

- Impact tests shall be performed at the test temperature specified by the Engineering Product Specification (EPS) and/or CMS-004.
- **CHINA CLOSED/OPEN DIE FORGINGS** SHALL HAVE ONE SET OF SPECIMENS TESTED PER THE EPS TEMPERATURE, ONE SET AT 0° F (-18° C) AND ONE SET AT THE MIDPOINT BETWEEN THE EPS TEMP AND 0° F (-18° C). IF EPS REQUIRES 0° F (-18° C) TEST THEN TEST A SECOND SET AT ROOM TEMPERATURE AND THE THIRD SET IS NOT NEEDED.
- A minimum of three impact specimens shall be tested to qualify a single heat of material.
- The results shall satisfy the requirements of the EPS and/or CMS-004.
- If the results of the impact tests are not satisfactory, then a retest shall be conducted using three specimens from the same location within the same QTC with no additional heat treatment. The results of retest shall meet or exceed the specific requirements.

	• Impact test of a QTC shall be used to qualify a heat and the bodies, bonnets, and end and outlet connectors produced from that heat.
Sub Size Specimens	When sub sized specimens are used the following adjustment factors shall be applicable to the specified sample sizes: **Dimension** / **AF**: 10 mm x 7.5 mm / 0.833; 10 mm x 5.0 mm / 0.667; 10 mm x 2.5 mm / 0.333
Hardness Testing	Heat treated material/material/part(s) and the QTC shall be hardness tested in accordance with procedures in ASTM E10 (Brinell) or ASTM E18 (Rockwell). Forgings may be Brinell hardness tested using portable hardness testers meeting the requirements of ASTM E110 or ASTM E10. At least two Rockwell or Brinell hardness test measurement shall be performed on the QTC after the QTC final heat treatment cycle. The QTC heat treatment cycles prior to hardness testing shall be the very same heat treatment cycles experienced by the tensile and impact test specimens.
Welding	Welding on the QTC is not allowed, except for attachment-type welds.
NDE	Any nondestructive examination specified by the EPS or the specific CMS shall be performed as the last operation prior to shipment and shall be performed in accordance with Cactus Wellhead NDE procedures or approved supplier NDE procedures.
Inspection	Material shall be inspected for injurious imperfections as follows: • Material in rough machined condition shall not exceed 250 AA in surface roughness in the machined areas and shall be free of burrs or injurious imperfections. • Material shall be free of internal forging bursts or hydrogen flakes and have a workmanlike finish. • Surface imperfections such as laps, seams, folds, tears, slivers, or mechanical marks, abrasions or pits need not be removed unless they infringe on the minimum wall thickness of the material or extend to a depth greater than 1/8 inch for forgings and 1/16 for bars.
Material Certification Report	Suppliers shall provide a Material Certification Report that meets the requirements of CMS-002 and the specific material CMS. Suppliers shall provide original manufacturer's certificate(s) that meet EN 10204 Type 3.1 requirements. All reports shall be furnished in electronic format and emailed to mtrs@cactuswellhead.com. In addition to the above, suppliers may also elect to provide the Cactus Wellhead Material Certification Report – Forgings, Bars, and Mill Shapes, Form CMS-002 as a tool to verify all required information is reported.

<table>
<tr>
<td>Material Certification Report Requirements</td>
<td>

Purchase Order No.& Line Item No.
Date
Packing List No.
Cactus Wellhead Material Specification (CMS)/Revision
CMS-002/Revision
Original mill/supplier identification
Testing Method Standards/Revision
Quantity
Part Number/Revision
Description/Size
Melt Practice
Forging Ratio
Material Grade
Grain Size for low alloy steel
Ideal Critical Diameter (Di) for 20” and larger forged bar
Heat Number
Heat Treat Lot Identification (as marked on material)

CHINA CLOSED/OPEN DIE FORGINGS MUST ALSO CONTAIN:

Oven load charts
Heat treatment plan
Photo of oven loading
All metallography photos at 100X and 500X magnification
Ferrite percentage when required by specific CMS
% Sheer and Lateral Expansion recorded on all Impact testing

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<td>

Serial Numbers for traceability to Nondestructive Examination (NDE) reports, when the Cactus Wellhead Purchase Order requires NDE.
Microstructure Photos when required by the specific CMS
Chemical Analysis
Mechanical Properties
Charpy Impact Properties, including temperature and Charpy values when required by the Cactus Wellhead Purchase Order
QTC Hardness (two readings)
QTC Size/Part
Actual heat treatment times and temperature for raw material and QTC
Start and finish temperatures for water and water based quenchants for quenched and tempered raw material.
Bar stock Hardness Profile for 20” and larger forged bar, three hardness readings required: OD, Center and Mid-Radius
No Weld Repair Statement
Material Capability Statement when required by the Cactus Wellhead Purchase Order
Furnace Calibration Statement stating that furnaces are calibrated and certified to API Spec 6A including Annex M for batch-type and SAE AMS2750 or SAE AMS-H-6875 for continuous-type.

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<td>Certificate of Conformance</td>
<td colspan="2">A Certification of compliance shall be provided. The COC shall state that the supplier certifies that the raw material has been tested and complies with the terms of the Cactus Wellhead Purchase Order and this CMS. The certificate shall be validated by an authorized supplier representative, independent of the manufacturing operations.</td>
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Prepared by:	Bryan Leach	Approved by:	Mat Trumbull
Date Prepared	October 18, 2024	Date Approved:	October 18, 2024

Revision History Table		
Rev L	10/18/2024	Added additional requirements for China Closed/Open die forgings.
Rev K	07/08/2021	Aligned standard sub-size specimens information with ASTM A370.
Rev J	07/29/2020	Updated references to include SAE AMS2750F, Pyrometry and SAE AMSH6875, Process for Heat Treatment of Steel. Updated Heat Treatment section for Equipment Qualification options. Updated to comply with API 6A 21st Edition Addendum 1.
Rev H	03/03/2020	Updated Material Certification Report section to require original manufacturer's certificate(s) that meet EN 10204 Type 3.1 to be provided.
Rev G	09/03/2019	Updated to comply with API 6A 21st edition.
Rev F	02/15/2018	Added to Reference Section: For undated references, the latest edition of the referenced document (including any amendments) applies, except that reference standards may be used upon issue and shall become mandatory six months from the date of the revision. Standardized on mostly ASTM testing method standards. Documentation must include the suppliers testing method standard and revision level.
Rev E	11/18/2015	Added all the reference industry standards used in throughout material specification to the Reference Section. Added a table to identify the applicable testing standards, ASTM, ISO, GB (Chinese). Added requirements to identify chemical, mechanical and impact test methods (updated Form CMS-002). Revised Certification of Compliance statement requirements to say the supplier certifies that the raw material has been tested and compliance with the terms of the Cactus Wellhead Purchase Order and this CMS.
Rev D	04/25/2012	Revised certification section to reflect the DI and hardness test profile fore 20" and greater forged bar. Expanded furnace calibration statement to include API Spec 6A Annex M for batch-type and SA AMS-H-6875 for continuous type.
Rev C	02/14/2012	Changes to Paragraph 5.6.4.1 and 5.7.4.1 in API 6A to reflect 12.55 mm (0.500 in) vs 12.7 mm (0.500) for the standard tensile specimen. Editorial change to correct an API error in conversion from Inches to millimeters. Added Form CMS-002, Material Certification Report as an option and revised certification requirements to align with the new form.
Rev B	10/18/2011	Added Grain size for low alloy steels and radioactivity free material to material processing. Added portable hardness tester option for hardness testing. Updated certification requirements.
Rev A	09/01/2011	Initial release